EXPERIMENT-3

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Course: BCA (2nd Semester)

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Academic Year: 2025-2026

Submitted To:

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sandeepsabharwal submit experiment 12ba7aa · 3 weeks ago

374 lines (374 loc) · 12.6 KB

Preview Code Blame Raw

1. Check whether a given number is divisible by 3 and 5 both

In [1]:

```
a = int(input("Enter a number"))
if a % 5 == 0 and a % 3 == 0:
    print("Divisible")
else:
    print("Non-Divisible")
```

```
Divisible
```

2. Check whether a given number is multiple of five or not

In [2]:

```
a = int(input("Enter a number"))
if a % 5 == 0:
    print("Multiple of 5")
else:
    print("Not a multiple of 5")
```

```
Multiple of 5
```

3. Find the greatest among two numbers. If numbers are equal than print “numbers are equal”.

In [4]:

```
a = int(input("Enter a number"))
b = int(input("Enter a number"))
if (a > b):
    print(a, "is greater")
elif (b > a):
    print(b, "is greater")
else:
    print(a, "and", b, "are equal")
```

```
5 and 5 are equal
```

4. Find the greatest among three numbers assuming no two values are same.

In [3]:

```
a = int(input("Enter a number"))
b = int(input("Enter a number"))
c = int(input("Enter a number"))
if (a == b) or (b == c) or (a == c):
    print("Two or Three values are same")
else:
    if (a > b) and (a > c):
        print(a, "is greater")
    elif (b > a) and (b > c):
        print(b, "is greater")
    else:
        print(c, "is greater")
```

```
Two or Three values are same
```

5. Check whether the quadratic equation has real roots or imaginary roots. Display the roots.

In [5]:

```
import math
a = int(input("Enter a number"))
b = int(input("Enter a number"))
c = int(input("Enter a number"))
D = (b * b) - 4 * a * c
if (D > 0):
    print("Two real and distict roots")
    root1 = (-b + math.sqrt(D)) / (2 * a)
    root2 = (-b - math.sqrt(D)) / (2 * a)
    print(root1, ",", root2, "are the roots")
elif (D == 0):
    print("Real and equal roots")
    root1 = (-b + math.sqrt(D)) / (2 * a)
    print(root1, "is the roots")
else:
    print("Imaginary roots")
    root1 = (-b + math.sqrt(D)) / (2 * a)
    root2 = (-b - math.sqrt(D)) / (2 * a)
    print(root1, ",", root2, "are the roots")
```

```
Real and equal roots
-2.0 is the roots
```

6. Find whether a given year is a leap year or not.

In [6]:

```
year = int(input("Enter any year"))
if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("Leap year")
else:
    print("Not leap year" )
```

```
Not leap year
```

7. Write a program which takes any date as input and display next date of the calendar

In [7]:

```
day = int(input("Enter day: "))
month = int(input("Enter month: "))
year = int(input("Enter year: "))
days_in_month = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31]
if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    days_in_month[1] = 29
day += 1
if day > days_in_month[month - 1]:
    day = 1
    month += 1
    if month > 12:
        month = 1
        year += 1
print("Next date is:", day, "/", month, "/", year)
```

```
Next date is: 5 / 3 / 2001
```

8. Print the grade sheet of a student for the given

8. Print the grade sheet of a student for the given range of cgpa. Scan marks of five subjects and calculate the percentage.

CGPA=percentage/10

CGPA range:

0 to 3.4 -> F

3.5 to 5.0->C+

5.1 to 6->B

6.1 to 7-> B+

7.1 to 8-> A

8.1 to 9->A+

9.1 to 10-> O (Outstanding)

Sample Gradesheet

Name: Rohit Sharma

Roll Number: R17234512 SAPID: 50005673

Sem: 1 Course: B.Tech. CSE AI&ML

Subject name: Marks

PDS: 70

Python: 80

Chemistry: 90

English: 60

Physics: 50

Percentage: 70%

CGPA:7.0

Grade: A

In [1]:

```
name = "Shreyansh Kumar Gupta"
r_num = "R010134"
sap_id = 590020790
sem = 1
course = "BCA"
c = 80
py = 85
web_tech = 100
maths = 80
evs = 100
per = (c + py + web_tech + maths + evs) / 5
```

```
per = (c + py + web_tech + maths + evs) / 5
cgpa = per / 10
if (0 <= cgpa <= 3.4):
    grade = "F"
elif (3.5 <= cgpa <= 5.0):
    grade = "C+"
elif (5.1 <= cgpa <= 6.0):
    grade = "B"
elif (6.1 <= cgpa <= 7.0):
    grade = "B+"
elif (7.1 <= cgpa <= 8.0):
    grade = "A"
elif (8.1 <= cgpa <= 9.0):
    grade = "A+"
elif (9.1 <= cgpa <= 10.0):
    cgpa = "O (Outstanding)"
else:
    grade = " "
print("Name:", name)
print("Roll number:", r_num, "\tSAPID:",sap_id)
print("Sem:", sem, "\tCourse:", course)
print()
print("Subject Name \t:\t Marks")
print("C \t\t:\t", c)
print("Python \t\t:\t", py)
print("Web Technology \t:\t", web_tech)
print("Maths \t\t:\t", maths)
print("EVS \t\t:\t", evs)
print("Percentage:", per)
print("CGPA:", cgpa)
print("Grade:", grade)
```

```
Name: Shreyansh Kumar Gupta
Roll number: R010134    SAPID: 590020790
Sem: 1  Course: BCA

Subject Name    :        Marks
```